

MUSTHOFA JOKO ANGGORO

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RESEARCH INTERESTS

Computer systems architecture, distributed systems, operating systems, hardware-software co-design — with particular interest in edge AI systems, resource-constrained computing, and energy-efficient system design across the computing continuum

EDUCATION

University of Indonesia (UI)

Bachelor of Computer Science, Information Systems

Jakarta, Indonesia

Expected Graduation: Aug 2026

• **GPA: 3.84/4.00**

- **Thesis Advisor:** Prof. Petrus Mursanto (Full Professor, former Dean of Faculty of Computer Science)
- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhanced Jump Branching Capabilities on Xilinx Spartan-3 FPGA (X3700AN) using VHDL
 - Extended baseline Morris Mano architecture by implementing jump branching capabilities in datapath, enabling more complex control flow operations
 - Designed and optimized digital logic circuits for hardware-level branching mechanisms, demonstrating proficiency in low-level system design and hardware-software co-design principles
 - Research informed by IEEE and ACM conference publications on computer architecture and FPGA-based system design
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS

INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee)

Singapore & Surabaya

July 2025

- Conducted sustainability-focused research on smart port automation and green technology adoption in distributed maritime systems, analyzing IoT sensor networks, automation architectures, and energy-efficient deployments
- Produced comprehensive technical report proposing tiered port automation framework emphasizing renewable energy utilization and carbon emission reduction — directly aligned with sustainability objectives in edge computing systems
- Collaborated with international research teams on applying systems thinking to environmental challenges

EXPERIENCE

Software Engineer Intern

Systems Engineering & Backend Development

Traveloka (Travel Technology Company)

Aug 2025 - Present

- **Edge AI-Relevant Skills:** Resource optimization, distributed system architecture, fault tolerance, cloud infrastructure (AWS, Terraform), containerization (Docker, Kubernetes), performance tuning, reliability engineering

SKILLS

Systems Programming: C, Python, Java, VHDL (hardware description), Go

Hardware & Architecture: FPGA development (Xilinx Spartan-3), VHDL, digital logic design, computer architecture implementation, hardware-software co-design

Machine Learning: Scikit-learn, TensorFlow, supervised/unsupervised learning, model evaluation, imbalanced learning

Distributed Systems: Microservices architecture, containerization (Docker, Kubernetes), cloud infrastructure (AWS, Terraform)

Operating Systems: Linux, Unix — OS internals, process management, memory management, system-level programming

RESEARCH & TECHNICAL PROJECTS

Used Car Sales Prediction System

Systems & ML Integration

Course Project - AI & Data Science

Apr 2025 - May 2025

- Designed end-to-end predictive analytics system integrating multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, K-Means clustering) with emphasis on data pipeline efficiency and model optimization
- Implemented systematic data preprocessing and feature engineering workflows to handle imbalanced datasets, demonstrating understanding of computational efficiency trade-offs in resource-constrained scenarios

- Applied performance evaluation methodologies and statistical analysis to optimize model selection and hyperparameter tuning
- **Technical Stack:** Python, Scikit-learn, Pandas, NumPy — with focus on algorithmic efficiency and scalable data processing